BSample Project Idea #1

Deep Learning for Kidney and Kidney Tumor Segmentation

Brief project description:

This collection contains CT scans and segmentations from subjects from the training set of the 2019 Kidney and Kidney Tumor Segmentation Challenge (KiTS19). The challenge aimed to accelerate progress in automatic 3D semantic segmentation by releasing a dataset of CT scans for 210 patients with manual semantic segmentations of the kidneys and tumors in the corticomedullary phase.

The imaging was collected during routine care of patients who were treated by either partial or radical nephrectomy at the University of Minnesota Medical Center. Many of the CT scans were acquired at referring institutions and are therefore heterogeneous in terms of scanner manufacturers and acquisition protocols. Semantic segmentations were performed by students under the supervision of an experienced urologic cancer surgeon.

***Dataset*:**

The dataset involved in this project can be downloaded from TCIA

https://wiki.cancerimagingarchive.net/pages/viewpage.action?pageId=61081171

Objective:

The objective is to develop deep learning methods to segment the kidney on the CT images and to segment the kidney tumors. More information can be found from the webpage above.